INCH-POUND
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COMMERCIAL ITEM DESCRIPTION

Calibrator, Torque-Wrench, 5 in-lbs – 2000 ft-lbs

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This document describes performance requirements for a torque-wrench calibrator and associated dead-weight calibration equipment.

2. CLASSIFICATION. The torque-wrench calibrator consists of two separate "stations." The two stations of the torque-wrench calibrator must be capable of being operated simultaneously. These systems are intended for use by civilian and military personnel to perform calibration of torque setting and indicating instruments, force gauges, dynamometers, tension-meters, and cable testers in aircraft and ground support equipment.

2.1. Order of precedence. In the event of a conflict between the text of this CID and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this CID shall take precedence. Nothing in this CID, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. SALIENT CHARACTERISTICS.

3.1. Item description. The torque-wrench calibrator is a high-accuracy instrument used to calibrate tools that apply a specific finite scalar force. This force causes a rotation (moment of force) about an axis on a tool, bolt, nut, or fastener to verify and adjust for the correct torque readings. The torque-wrench calibration system shall have two stations; a low-torque and a high-torque station. The measurements shall be displayed in the United States customary system and the National Institute of Standards and Technology (NIST) SP811, *Le Systéme International d´Unités* (SI), as chosen by the user.

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AMSC: N/A FSC 6635

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3.2. Calibrator type. The torque-wrench calibrator is a dead-weight, bench type with two stations; a low-torque (manual) station for calibration of torque wrenches (spring, torsional-element, ratcheting, and beam-type) and torque screwdrivers.

3.3. Calibrator torque range. The torque-wrench calibrator shall have a low-range station with a fully-variable, bi-directional torque-calibrating range from 5 inch-pounds (in-lb) to 250 foot-pounds (ft-lbs), and a second station with a fully-variable, bi-directional, electro-mechanically generated torque and force calibration range from 30 in-lbs to 2000 ft-lbs.

TABLE. Torque range conversion/equivalence.

Equivalent Torque Measurements						
Torque Magnitude	In-oz	Ft-Lbs	N-m	N-cm	m-kg	dN-m
5 In-oz	–	0.0260	0.0353	3.53	0.0036	3.53
30 In-oz	–	0.1563	0.2118	21.18	0.0216	2.118
250 Ft-lbs	47999.99	–	338.954	33895.44	34.56	3389.54
2000 Ft-lbs	383999.99	–	2711.636	271163.58	276.51	27116.36

3.4. Calibrator low-end torque accuracy. The torque-wrench calibrator shall have a low-range station with a fully-variable, torque calibrating range from 5 in-lbs to 250 ft-lbs with a minimum accuracy of ±0.25%, and a precision repeatability ≥ ±99% @ 25 °C.

3.5. Calibrator high-end torque accuracy. The torque-wrench calibrator shall have a high-range station with a fully-variable, torque calibrating range from 30 in-lbs to 2000 ft-lbs with a minimum accuracy of ±0.25%, and a precision repeatability ≥ ±99% @ 25 °C.

3.6. Calibrator torque variability. The torque-wrench calibrator shall have a measurement variability that includes the combined effects of non-linearity, hysteresis, zero drift, creep, temperature coefficient, and any other error contributions ≤ ±1.0% of full-scale measurement/reading.

3.7. Calibrator torque transducers. The torque-wrench calibrator shall have a set of several torque transducers capable of transmitting the exerted rotational force and span the range of applied moments of force within the accuracy and precision limits.

3.8. Torque transducer span. The torque-wrench calibrator transducers shall have an over-span from 5 to 10% into the next range of the applied moment of force and perform within the accuracy and precision limits.

3.9. Calibrator torque transducer positioning. The torque-wrench calibrator shall have the torque transducers positioned or aligned on a plane perpendicular or parallel to the working surface based on the user's needs and equipment being calibrated.

3.10. Calibrator over-torque safety limit. The torque-wrench calibrator shall have a safety feature that restricts or limits maximum creep, measured from the initial reading on loading to the reading after five minutes, must not exceed 0.15% of full scale for each transducer.

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3.11. Torque calibrator drive fixtures. The torque-wrench calibrator shall have a "dummy-drive" socket base or "passive-reaction post" for the operator to exercise torque wrenches prior to mounting on the calibrator. The "dummy" square-drive sockets (sizes ¼-inch, ⅜-inch, ½-inch, ¾-inch, and 1-inch female square) shall be positioned in a location convenient to the operator.

3.12. Torque calibrator drive adapters. The torque-wrench calibrator shall have a set of drive socket adapters for square-drive sockets (sizes ¼-inch, ⅜-inch, ½-inch, ¾-inch, and 1-inch female square), hex-drive torque screwdrivers, and any other adapters which shall be required to calibrate the test item. These adapters shall not, when placed in the system, have free rotational play greater than 1°. A means of compensation shall be provided to keep the moment arm at 90° ±1 with any applied weight, i.e., the calibration method must ensure no cosine error is introduced to the transducer during calibration. All drive adapters shall conform to specifications of ASME B107.300.

3.13. Calibrator torque joint-rate simulator. The torque-wrench calibrator shall have a joint-rate simulator adapter(s) with sizes to fit ¼-inch, ⅜-inch, ½-inch, ¾-inch, and 1-inch wrench and power tool "square drives" or "shanks," either "tapered" or "straight" to test and calibrate power tools (electric, mechanical, or pneumatic), excluding impact or pulsed-energy tools.

3.14. Torque calibrator work plane. The torque-wrench calibrator reaction post must "glide" or "slide" on a horizontal guide with quick-adjust settings and have a maximum possible misalignment of 3° on any plane, measured from the transducer platform to the free-standing instrument.

3.15. Torque calibrator stability. The torque-wrench calibrator shall be a work-station system with a static and secure surface that will not allow for bending, buckling, vibration, sliding, torsion, or show effects of fatigue while performing the calibration functions.

3.16. Digital display indicator. Each section of the torque-wrench calibrator shall have an electronic, digital readout display(s) capable of displaying all the information related to the "Power On" self-test, tool test, measured torque, and instrument calibration. At a minimum, the torque readings shall be displayed in ≥ 0.5-inch character/digits in height.

3.17. Control panel. The torque-wrench calibrator control functions shall be operable using the front panel controls. The labels and indicators, controls, and switches shall provide for maximum intelligibility.

3.18. Visual indicators. The torque-wrench calibrator shall have aural and/or visual indicator(s), power on/off, applicable engineering units, transducer in-use range, torque over-range (greater than 100% of transducer range), first Peak-of-Torque, peak, track, and clockwise or counterclockwise operation.

3.19. Power "On" self-test. The torque-wrench calibrator unit shall have a (self-contained) power "On" self-test (POST) and dynamic diagnostics on demand by the user. The POST normally tests power, interrupts, reference, oscillator/exciter, temperature (compensation), strain gauge (torque sensor), Gauge Bridge, shunt calibration, comparator, amplifier, A/D converter, display, and display driver.

3.20. Auto-zero function. The torque-wrench calibrator unit shall reach an "auto-zero" condition after a "power on" and successful "POST."

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3.21. Calibration traceability/compliance. The torque-wrench calibrator unit shall have tolerance standards (physical/dimensional) and traceability in accordance with the NIST, Technical Note 1297 [6].

3.22. Calibration period. The design of the torque-wrench calibrator shall provide readily-accessible calibration and maintenance adjustments. The calibration adjustments shall be accessible without removal of the test-set case or modules. The calibration interval shall be based on a period of one calendar year (8760 hrs.) or an operating time of 2,080 hours, whichever occurs first. At the end of this interval, a minimum of 85% of the units shall remain in tolerance in accordance with MIL-PRF-28800, paragraph 3.12.

3.23. Non-operating temperature. The torque-wrench calibrator unit shall be subjected to a non-operating temperature in the range from -20 to 50 °C (-2 to 122 °F).

3.24. Operating temperature. The torque-wrench calibrator unit shall be subjected to an operating temperature in the range from 10 to 32 °C (50 to 90 °F).

3.25. Humidity. The torque-wrench calibrator unit shall be capable of operating in a relative humidity (non-condensing) of 5 to 90% ±1% in the temperature range of 10 to 32 °C (50 to 90 °F).

3.26. Power requirements. The torque-wrench calibrator shall have 120/220 VAC ±10%, 50/60-Hz nominal power requirements. A detachable power connector(s) shall be provided with the system. These connectors shall be a NEMA 5-15 style standard plug.

3.27. Power connections. The torque-wrench calibrator shall have a power cord with a length ≥5 ft.

3.28. Torque transducer power connections. The torque-wrench calibrator shall have detachable cables provided for each torque transducer in the low torque section and at least one 5 ft detachable cable shall be provided with the high torque section. Each section shall include one spare cable.

3.29. Printer (Hardcopy). The torque-wrench calibrator shall have hardcopy printing functionality through an RS-232, DB-9P, serial communications port. A provided printing device shall have a minimum 16- character width and two-line-per-second speed. The printer may be either internal to the digital indicator or in a separate enclosure, and be physically and electrically attached to the digital indicator. The printer shall be capable of printing torque readings and units directly from the digital indicator. A “PRINT” button shall be provided on either the indicator or printer. A paper feed button shall also be provided. The printer shall be provided with a device for allowing the operator to easily tear off a length of paper. At least six rolls of printer paper shall be provided, as well as three spare ink sources (ribbon, toner cartridge, ink cartridge, or other technology). The original manufacturer’s part number for the printer shall be noted in the equipment manual. The paper and ink source shall be of a commercially-available type. The paper size and ink cartridge manufacturer and part number shall be noted in the equipment manual.

3.30. System cabinet. The torque-wrench calibrator shall be supplied with a heavy-duty storage cabinet. The cabinet shall accommodate all adapters and fixtures (with the exception of calibration weights and moment arms) in an orderly and accessible manner. Color and finish of the storage cabinet shall be identical to that of the main instrument. Knobs or handles on the cabinet shall not interfere with the operation of the mechanical loader. The storage cabinet shall

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store the low-torque calibration section when those instruments are not in use. The cabinet shall have casters with brakes that can be locked during use or storage.

3.31. Design and construction. The torque-wrench calibrator shall be designed and constructed in accordance with all applicable European Union (EU) requirements with the "CE" marking affixed, International Organization for Standardization (ISO), the International Electrotechnical Commission (IEC), the International Telecommunication Union (ITU), and the American National Standards Institute (ANSI). The torque-wrench calibrator shall be compliant with the Restriction of Use of Hazardous Substances (RoHS) and Waste from Electrical and Electronic Equipment (WEEE) guidelines and policies. It shall be designed and constructed so that no parts will work loose in service, and to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.32. Maintainability. The torque-wrench calibrator unit shall meet the maintainability criterion in accordance with MIL-STD-1472, paragraphs 5.9 through 5.9.18 and 5.9.21, and MIL-PRF-28800, paragraphs 3.14.1 through 3.14.4, 6.6.2.4, and 6.6.2.4.1.

3.33. Preventive maintenance. Preventive maintenance shall not require more than 15 minutes per 30-day period. Preventive maintenance shall not require breaking of the equipment seams where calibration seals would normally be placed in accordance with MIL-PRF-28800, paragraphs 3.14.2 and 3.14.3.

3.34. Corrective maintenance. The torque-wrench calibrator unit shall be designed to meet reparability criterion in accordance with MIL-PRF-28800, paragraphs 3.14.3 and 3.14.3a through d.

4. REGULATORY REQUIREMENTS.

4.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life-cycle costs. The offeror/contractor is encouraged to use recovered materials or biobased products to the maximum extent practicable, in accordance with Federal Acquisition Regulation (FAR), part 23.403. However, used, rebuilt, or refurbished items shall not be provided.

4.2 Green Procurement Program. Green Procurement Program (GPP) is a mandatory federal acquisition program that focuses on the purchase and use of environmentally preferable products and services. GPP requirements apply to all acquisitions using appropriated funds, including services and new requirements. FAR 23.404(b) applies and states the GPP requires 100 percent of EPA-designated product purchase that are included in the Comprehensive Procurement Guidelines list that contains recovered materials, unless the item cannot be acquired: a) competitively within a reasonable timeframe; b) meeting reasonable performance standards; or c) at a reasonable price. The prime contractor is responsible for ensuring that all subcontractors comply with this requirement.

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5. PRODUCT CONFORMANCE. Verification of the delivered torque-wrench calibrator unit set will be performed by the end user at the delivery point. Misalignment of components; conditions that present a safety hazard to personnel during operation, servicing, or maintenance; and evidence of corrosion, deterioration, or inability to calibrate or service will be cause for rejection.

6. PACKAGING. Preservation, packing, and markings shall be as specified in the contract or order.

7. NOTES.

7.1 Source of documents.

7.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks listed below form a part of this specification to the extent specified herein.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-28800	-	General Specification for Test Equipment for use with Electrical and Electronic Equipment
MIL-STD-1686	-	Electrostatic Discharge Control Program for Protection of Electrical and Electronic Parts, Assemblies and Equipment (excluding electrically initiated explosive devices)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1472	-	Human Engineering
MIL-STD-461	-	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment

(Copies of these documents are available at https://assist.dla.mil.)

7.1.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B107.300 - 2010	-	Torque instruments

(Copies of these documents are available at https://www.asme.org.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E2428 - 15a	-	Standard Practice for Calibration and Verification of Torque Transducers

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ASTM E2624 - 17	-	Standard Practice for Torque Calibration of Testing Machines

(Copies of these documents are available at http://www.astm.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 6789-1	-	Assembly tools for screws and nuts – Hand torque tools - Requirements and methods for design conformance testing and quality conformance testing; minimum requirements for declaration of conformance

(Copies of these documents are available at http://www.astm.org.)

INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

IEC 60529 (Corr 2)	-	Degrees of Protection Provided by Enclosures (IP Code) - Edition 2.2 Consolidated Re-print

(Copies of these documents are available at http://www.iec.ch.)

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY (NIST)

NIST SP 811	-	Guide for the Use of the International System of Units (SI)
NIST Technical Note 1297	-	Guidelines for Evaluating and Expressing the Uncertainty of NIST Measurement Results

(Copies of these documents are available at https://www.nist.gov.)

7.2 Ordering data. The contract or order should specify the following:

a. Title, number, and date of this CID.

7.3 Key words.

Dual station usage
Dynamometers Force
Gages

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